DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Supplementary Examination – Summer 2022

Course: B. Tech. **Branch :** Electronics and Telecommunications

Semester : 5

Subject Code & Name: BTEXPE506D Introduction to MEMS

Max Marks: 60 **Date:** **Duration: 3 Hr.**

Instructions to the Students:

1. *All the questions are compulsory.*
2. *The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in ( ) in front of the question.*
3. *Use of non-programmable scientific calculators is allowed.*
4. *Assume suitable data wherever necessary and mention it clearly.*

		(Level/CO)	Marks
Q. 1	**Solve Any Two of the following.**		
A)	Explain in detail Principles, application and design of MEMS.	**L1/CO1**	**6**
B)	Explain in detail Analog control of MEMS	**L1/CO2**	**6**
C)	What is MEMS fabrication modules.	**L1/CO3**	**6**
Q.2	**Solve Any Two of the following.**		
A)	Explain in detail wet and dry etching processes of micromachining.	**L2/CO1**	**8**
B)	Explain in detail Materials for MEMS of Substrate and wafers.	**L2/CO2**	**8**
C)	What is role of Oxidation in MEMS	**L2/CO3**	**8**
Q. 3	**Solve Any One of the following.**)		
A)	Explain following points 1. Accelerometers 2.gyroscopes	**L3/CO1**	**10**
B)	Explain Active substrate material with piezoelectric and Lithography (LIGA).	**L3/CO2**	**10**
Q.4	**Solve Any Two of the following.**		
A)	Explain and types of Surface Micromachining	**L1/CO5**	**6**
B)	Explain in detail Mechanics of solids in MEMS	**L1/CO4**	**6**
C)	Explain in detail , Hookes‟s law.	**L1/CO5**	**6**
Q. 5	**Solve Any One of the following**		
A)	Explain in detail Isotropic Etching process and Anisotropic Etching	**L1/CO4**	**10**
B)	Expalin Modeling of Coupled Electromechanical Systems.	**L1/CO6**	**10**

***** End *****

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